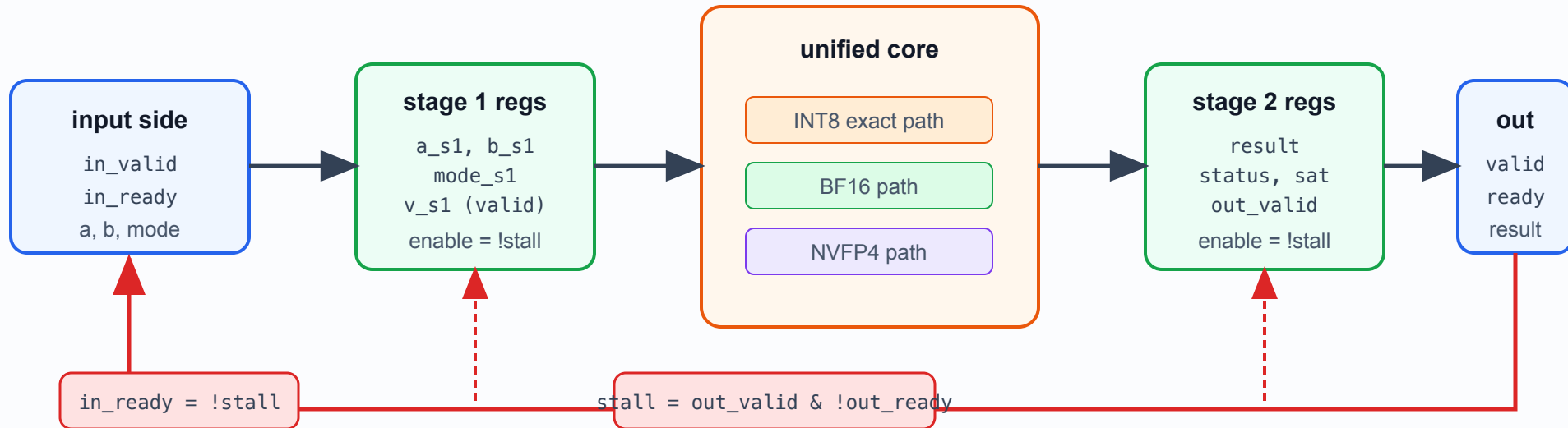


dotprod_seq Ready/Valid Wrapper

Two register stages around the unified INT8/BF16/NVFP4 combinational core

valid pipeline (2-deep): $\text{in_valid} \ \& \ \text{in_ready} \rightarrow \text{v_s1} \rightarrow \text{out_valid}$



One stall (a single always_ff enable) freezes both stages together: no drop, no reorder, no partial update.

Not an FSM: control is one valid bit shifting down the pipe, not a state graph. Formal focus: held output stays valid and stable while `out_valid` is high and `out_ready` is low.